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THE Q_p-FOLIATION INTERFERENCE THEOREM: LEADING-DIGIT EXPONENTIAL SUMS ON p-ADIC SCHOTTKY GROUPS

Main result: the additive-character sum of leading p-adic digits over closed non-backtracking walks on a Q_p-Foliation is a Gauss-sum interference of $p - 1$ multiplicative sectors; it has Hankel rank exactly $p - 1$, so classical transfer matrices need dimension $\geq p - 1$, while the quantum layer evaluates it on $\lceil \log_2 p \rceil$ qubits with $O(1/\varepsilon)$ queries.

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Abstract

We formulate a two-layer framework, the $\mathcal{Q}_p$-Foliation, coupling a p-adic layer (a non-Archimedean local system induced by a Schottky group $\Gamma \subset PGL_2(\mathbb{Q}_p)$ in good position) with a quantum layer (a digit-transport register carrying $\mathbb{C}[\mathbb{F}_p^*]$). The p-adic layer yields a product formula for the leading p-adic digit of the dominant eigenvalue of every closed non-backtracking walk, uniform in word length. On the leaves of the valuation foliation we define the leading-digit exponential sum $S_n = \Sigma\ \psi(\bar{a}(\delta))$ and prove: (i) S_n decomposes as a Gauss-sum-weighted interference of $p - 1$ twisted non-backtracking traces, each of which factors along the walk while S_n does not; (ii) the associated path function has Hankel rank exactly $p - 1$, so every classical transfer-matrix representation requires dimension at least $p - 1$; (iii) the quantum layer estimates S_n/N_n to additive error ε with $O(1/\varepsilon)$ calls to a circuit of size $O(n(\log d + M(\log p)))$ on $n\lceil\log_2 d\rceil + \lceil\log_2 p\rceil$ qubits. We state precisely which classical methods are separated and which are not. See Appendix A for Validation Code access.

1. Introduction and scope

Walk algorithms on quotients of the Bruhat–Tits tree usually compute quantities that factor along the walk: a product of local weights, evaluated by a small transfer matrix. Such quantities are classically cheap, and no quantum layer can improve on them in a meaningful way. The purpose of this paper is to isolate a number-theoretic walk invariant that provably does not factor, and to show that the quantum layer of a $\mathcal{Q}_p$-Foliation evaluates it with resources that transfer matrices cannot match.

The mechanism is elementary to state. Leading p-adic digits multiply along a walk (Theorem 1). A multiplicative character of $\mathbb{F}_p^*$ therefore factors along the walk; an additive character does not. Its expansion in multiplicative characters has $p - 1$ nonzero Gauss-sum coefficients, and this is exactly what forces transfer matrices of dimension $p - 1$ while a quantum register of $\lceil\log_2 p\rceil$ qubits holds all sectors in superposition.

Scope of the claims. The separation proved here is (a) unconditional against classical transfer-matrix evaluation and (b) quadratic in precision against classical sampling estimators. We do not claim a separation from all classical algorithms; proving such a statement would separate classical from quantum computation, which is currently out of reach. Section 7 states what remains open.

2. The $\mathcal{Q}_p$-Foliation framework

2.1 The p-adic layer

Let $\Gamma \subset PGL_2(\mathbb{Q}_p)$ be a Schottky group of rank $g \geq 2$ with free generators $g_1, \ldots, g_g$, acting on the Bruhat–Tits tree $\mathcal{T}_p$. Write $X = \mathcal{T}_p/\Gamma$ for the quotient core, taken to be the rose with $d = 2g$ oriented edges (the letters $g_i^{\pm 1}$). Each letter e has an attracting fixed point a_e, a repelling fixed point r_e, and a multiplier $\mu(e) \in p\mathbb{Z}_p$, so that $a_{\bar{e}} = r_e$ and $r_{\bar{e}} = a_e$.

Definition 2.1 ($\mathcal{Q}_p$-Foliation, p-adic layer). A $\mathcal{Q}_p$-Foliation over X is a triple $\mathcal{F} = (V, \{\nabla_e\}, \mu)$ where $V_v \cong \mathbb{Q}_p{}^2$ carries the sup-norm at each vertex, $\nabla_e \in GL_2(\mathbb{Q}_p)$ is the normalized generator matrix of e, and $\mu(e)$ is its multiplier, with $|\mu(e)|_p \leq p^{-t}$ for a fixed $t \geq 1$. It satisfies the following axioms.

Axiom	Statement
I. Reversal	$\nabla_{\bar{e}} = \nabla_e^{-1}$.
II. Ultrametric sub-additivity	$\|x + y\| \le \max(\|x\|, \|y\|)$ on V_v. This is a property of the sup-norm on $\mathbb{Q}_p^2$, hence automatic.
III. Contraction	$\nabla_e = u_e v_e^T + \mu(e) R_e$, with $v_e^T u_e = 1$ and $\|R_e\| \le 1$. Here $u_e = (a_e, 1)^T$ and $v_e^T = (1, -r_e)/(a_e - r_e)$. This is derived from ping-pong by diagonalizing g_e at its fixed points.
IV. Good position	For every allowed junction $e \to e'$ ($e' \neq \bar{e}$), the junction weight $w(e, e') = v_e^T u_{e'} = (a_{e'} - r_e)/(a_e - r_e)$ is a p-adic unit.

Remark 2.2. A sufficient condition for Axiom IV is that the 2g fixed points lie in 2g distinct residue classes modulo p (all ping-pong discs are maximal balls of equal radius). This requires $p \ge 2g$.

2.2 Theorem 1: the leading-digit product formula

For a cyclically reduced word $\delta = (e_1, \ldots, e_n)$, indices mod n, let $r(\delta)$ be the dominant eigenvalue of $\nabla_{e_n} \cdots \nabla_{e_1}$.

Theorem 1 (Product formula). Under Axioms I–IV,

$$r(\delta) = \prod_{k=1}^{n} w(e_k, e_{k+1}) \cdot (1 + O(p^{-t})),$$

uniformly in n. In particular the valuations add exactly, and the leading p-adic digit of $r(\delta)$ is

$$\bar{a}(\delta) = \prod_{k=1}^{n} a(e_k, e_{k+1}) \bmod p, \quad a(e, e') := \text{leading digit of } w(e, e') \in \mathbb{F}_p^*.$$

Proof. Expand $\prod_k (u_k v_k^T + \mu_k R_k)$. The rank-one string equals $u_1 \cdot \prod_k (v_k^T u_{k+1}) \cdot v_n^T$. Every other term contains at least one factor $\mu_k R_k$, so by Axiom II the remainder is bounded by the largest such term, of relative size at most p^{-t} by Axiom IV, independently of n. The cyclic trace of the rank-one string is $\prod_k w(e_k, e_{k+1})$. Since det = $\prod \mu_k$, the second eigenvalue is $\det/\lambda_1$, which is smaller than λ_1 by a factor of at least p^{-nt}; hence λ_1 equals the trace up to relative error p^{-t}. ■

2.3 Leaves

The valuation foliation of walk space has leaves $L_m = \{\delta : \text{val det } \gamma_\delta = m\}$. Under uniform contraction t, the leaf L_{nt} is exactly the set C_n of cyclically reduced words of length n. Distinct words in a common leaf have equal valuation; these are the words whose contributions interfere in Section 5.

2.4 The quantum layer

Axiom	Statement
V. Digit transport	The fiber is $\mathcal{H} = \mathbb{C}[\mathbb{F}_p^*]$. Each allowed junction $e \to e'$ acts by the unitary permutation $\|y\rangle \mapsto \|a(e, e') \cdot y\rangle$.
VI. Leaf interference	The leaf observable is the additive-character phase $\Phi\|y\rangle = \psi(y)\|y\rangle$, with $\psi(y) = e^{2\pi i y/p}$.

3. Notation

- χ ranges over the $p-1$ multiplicative characters of $\mathbb{F}_p^*$; 1 denotes the trivial character.
- $G(\chi) = \Sigma_{z\in\mathbb{F}_p^*} \chi(z)\,\psi(z)$ is the Gauss sum. Then $|G(\chi)| = \sqrt{p}$ for $\chi \neq 1$ and $G(1) = -1$.
- B_χ is the twisted non-backtracking (Hashimoto) matrix on oriented edges: $(B_\chi)_{e,e'} = \chi(a(e, e'))$ if $e' \neq \bar{e}$, and 0 otherwise. B_1 is the ordinary Hashimoto matrix.
- $N_n = |C_n| = \mathrm{Tr}(B_1^n) = (2g-1)^n + g + (g-1)(-1)^n$.
- $M(b)$ is the cost of multiplying two b-bit integers.

4. Main theorem

Define the leading-digit exponential sum on the leaf L_{nt}:

$$S_n = \Sigma_{\delta\in C_n} \psi(\bar{a}(\delta)).$$

Hypothesis (H). For some allowed junction (e^*, e'), the digit products of non-backtracking prefixes ending at e^* cover $\mathbb{F}_p^*$, and so do those of non-backtracking suffixes starting at e'.

Theorem 2 ($\mathcal{Q}_p$-Foliation Interference Theorem). Let $\mathcal{F}$ be a $\mathcal{Q}_p$-Foliation satisfying Axioms I–VI with uniform contraction t, and assume (H). Then:

(1) **Sector decomposition.** $S_n = \Sigma_\chi [G(\bar{\chi})/(p-1)] \cdot \mathrm{Tr}(B_\chi^n)$. Every sector factors along the walk; S_n is an interference of all $p-1$ sectors with weights of modulus $\sqrt{p}/(p-1)$ (and $1/(p-1)$ for $\chi = 1$), none of them zero.

(2) **Classical transfer matrices cannot factor it.** The path function $\varphi = \psi \circ \bar{a}$ has Hankel rank exactly $p-1$. Every transfer-matrix representation over $\mathbb{C}$ that evaluates φ edge by edge has dimension at least $p-1$. By contrast, $\chi \circ \bar{a}$ factors with one scalar per junction.

(3) **The quantum layer evaluates it.** Using $n\lceil\log_2 d\rceil + \lceil\log_2 p\rceil$ qubits, the quantum layer estimates S_n/N_n to additive error ε with probability at least $2/3$, using $O(1/\varepsilon)$ calls to a circuit of size $O(n(\log d + M(\log p)))$.

(4) **Separation.** Transfer-matrix evaluation requires dimension $\Theta(p)$; classical sampling estimators require $\Theta(1/\varepsilon^2)$ samples; the quantum layer requires $O(1/\varepsilon) \cdot \mathrm{poly}(n, \log p)$. At the square-root cancellation scale $\varepsilon \approx N_n^{-1/2}$, this is $\Theta(\min(N_n, p))$ classically against $O(\sqrt{N_n})$ quantumly.

5. Proof

5.1 Sector decomposition

Expand the additive character in multiplicative characters. For $z \in \mathbb{F}_p^*$,

$$\psi(z) = \Sigma_\chi\, \hat{c}(\chi)\, \chi(z), \quad \hat{c}(\chi) = (1/(p-1))\, \Sigma_z\, \psi(z)\, \bar{\chi}(z) = G(\bar{\chi})/(p-1).$$

Because χ is multiplicative and $\bar{a}(\delta)$ is a product of junction digits (Theorem 1), $\chi(\bar{a}(\delta)) = \prod_k \chi(a(e_k, e_{k+1}))$. Summing over closed non-backtracking walks of length n is exactly the trace of the n-th power of the twisted Hashimoto matrix:

$$\Sigma_{\delta \in C_n} \chi(\bar{a}(\delta)) = \mathrm{Tr}(B_\chi^n), \quad \text{hence} \quad S_n = \Sigma_\chi \, [G(\bar{\chi})/(p-1)] \cdot \mathrm{Tr}(B_\chi^n).$$

The Gauss-sum moduli $|G(\bar{\chi})| = \sqrt{p}$ ($\chi \neq 1$) and $|G(1)| = 1$ show that no coefficient vanishes. ■

5.2 Classical transfer matrices cannot factor it

Extend φ to all words over the edge alphabet by $\varphi(e_1 \cdots e_n) = \psi(\prod_{k=1}^{n-1} a(e_k, e_{k+1}))$ on non-backtracking words and 0 otherwise. Its Hankel matrix is $H[u, v] = \varphi(uv)$.

1. Fix the junction (e^*, e') of Hypothesis (H) and put $c = a(e^*, e')$. For prefixes u ending at e^* and suffixes v starting at e', $\varphi(uv) = \psi(c \cdot a(u) \cdot a(v))$.
2. By (H), as u and v vary, a(u) and a(v) each cover $\mathbb{F}_p^*$. So H contains the $(p-1) \times (p-1)$ block $K[x, y] = \psi(cxy)$, $x, y \in \mathbb{F}_p^*$.
3. By 5.1, $K = \Sigma_\chi \, \hat{c}(\chi) \, \chi(c) \cdot (\chi \otimes \chi)$. This is a sum of $p-1$ rank-one matrices built from linearly independent (orthogonal) character vectors, with every coefficient nonzero. Hence rank $K = p-1$, and rank $H \geq p-1$. The matching upper bound comes from the explicit representation of dimension $d(p-1)$ on edges × digits.
4. By the Fliess / Carlyle–Paz theorem, the minimal dimension of a weighted automaton (a product of transfer matrices) computing a word function equals the rank of its Hankel matrix. Therefore every edge-by-edge transfer-matrix evaluation of φ has dimension at least $p-1$.
5. For a multiplicative character, the same block is $\chi(c) \cdot \chi(x) \, \chi(y)$, which has rank 1: $\chi \circ \bar{a}$ factors with one scalar per junction, as does the leading digit of Theorem 1 itself.

This is the precise sense of "does not factor along the walk": multiplicative characters of the leading digit factor; additive characters cannot. ■

5.3 The quantum layer evaluates it

1. **Prepare words.** Load the word register into the uniform superposition over reduced words of length n: d choices for the first letter, then $d-1$ non-backtracking choices for each subsequent letter. This uses $n\lceil \log_2 d \rceil$ qubits and $O(n \log d)$ gates.
2. **Flag closure.** Compute the flag $f = [e_n \neq \bar{e}_1]$ into an ancilla; it marks the cyclically reduced words C_n.
3. **Transport digits (Axiom V).** Initialize the digit register to $|1\rangle$ and multiply it by $a(e_k, e_{k+1})$ at each junction, including the closing junction on flagged words. Each step is a reversible modular multiplication by a table constant, cost $O(M(\log p))$.
4. **Interfere (Axiom VI).** A Hadamard test with the controlled phase Φ applied on the flagged branch returns the real and imaginary parts of $E[f \cdot \psi(\bar{a})]$. Garbage registers coincide in both branches and do not affect the test. Within each branch, distinct words of equal valuation contribute with phases $\psi(\bar{a}(\delta))$; this is the interference of Theorem 2(1), with all $p-1$ sectors carried by $\lceil \log_2 p \rceil$ qubits.
5. **Normalize.** The flag probability is exactly $N_n/(d(d-1)^{n-1})$, bounded below by a constant for $d \geq 4$. Dividing by it gives S_n/N_n.
6. **Amplify.** Amplitude estimation applied to the Hadamard test reaches additive error ε with probability at least $2/3$ using $O(1/\varepsilon)$ calls to the circuit. ■

5.4 Separation

The transfer-matrix bound is 5.2 together with the matching upper bound $d(p-1)$. The sampling bound is the standard lower bound for estimating the mean of a bounded random variable to $\pm\varepsilon$, which requires $\Omega(1/\varepsilon^2)$ samples in the worst case; amplitude estimation is optimal at $O(1/\varepsilon)$. At $\varepsilon \approx N_n^{-1/2}$, classical sampling is no cheaper than enumerating C_n, so the classical cost is $\Theta(\min(N_n, p))$ up to polynomial factors, against $O(\sqrt{N_n})$ for the quantum layer. ■

Method	**Resource**	**Cost**
Transfer matrices	State dimension	$\Theta(p)$; time $\Theta(n\,p\,d)$
Classical sampling	Samples	$\Theta(1/\varepsilon^2) \cdot \mathrm{poly}(n, \log p)$
$\mathcal{Q}_p$-Foliation quantum layer	Circuit calls	**$O(1/\varepsilon) \cdot \mathrm{poly}(n, \log p)$**
All three at $\varepsilon \approx N_n^{-1/2}$	Total	$\Theta(\min(N_n, p))$ classical vs $O(\sqrt{N_n})$ quantum

6. Interpretation

Theorem 1 shows that the p-adic layer reduces every closed walk to a multiplicative digit. Theorem 2 shows that the number-theoretically natural observable on that digit, the additive character, is exactly the one that escapes multiplicative structure. Classically, the $p-1$ Gauss-sum sectors must be held as $p-1$ explicit dimensions; in the quantum layer they coexist in one register of $\lceil \log_2 p \rceil$ qubits, and their interference is read out directly.

7. Scope and open problems

- **Not claimed.** A quantum advantage over every classical algorithm. A cleverer classical method below cost $\Theta(p)$ is not excluded; for example, when $p-1$ is smooth, discrete logarithms are cheap and the sector decomposition is classically explicit. Ruling such methods out is the central open problem.
- **Regime of interest.** The separation is strongest for cryptographically large p, at target precision near the square-root cancellation scale $N_n^{-1/2}$.
- **Non-uniform contraction.** With generator-dependent t_e, adjoin a valuation-clock register so that the quantum layer sums over a leaf L_m rather than over word length.
- **Higher digits.** Controlling carries from shorter words would extend Theorem 1 beyond the leading digit and give n-th digits of the Schottky period.
- **Novelty.** The individual tools (Schottky ping-pong, Gauss sums, Hankel rank, amplitude estimation) are classical. Whether the combination and the invariant S_n appear in prior literature has not yet been checked.

Appendix A. Validation Code

Every dimension and variance claim in this paper can be checked independently with the open validation code released with it under the MIT License:

Repository: https://github.com/teknoquants/THE-p-FOLIATION-INTERFERENCE-THEOREM-LEADING-DIGIT-EXPONENTIAL-SUMS-ON-p-ADIC-SCHOTTKY-GROUPS

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